

# Dramane Diarra, Ph.D.

**6G/AI/ML Researcher | Wireless Communications | AI-Driven Network Optimization**  
[dramanediarra007@gmail.com](mailto:dramanediarra007@gmail.com) | [linkedin.com/in/drdiarra](https://linkedin.com/in/drdiarra) | +254 746 885 584 / +223 75 04 54 43  
Nairobi, 00200, Kenya | [drdiarra.github.io/DrD-portfolio](https://drdiarra.github.io/DrD-portfolio)



## RESEARCH PROFILE

Ph.D. in Electrical Engineering (Telecommunications) with research expertise at the intersection of Artificial Intelligence, Reinforcement Learning, wireless communications and network optimization. Experienced in developing AI-driven solutions for wireless network problems, with particular emphasis on Deep Reinforcement Learning, Multi-Agent Reinforcement Learning, intelligent resource allocation, power control, energy-efficient network optimization and large-scale cell-free massive MIMO systems. Strong hands-on experience with Python, PyTorch, TensorFlow/Keras, MATLAB and machine learning workflows, including model development, training, evaluation, numerical simulation, optimization and large-scale computational experimentation. Research experience covers 5G/6G wireless networks, mmWave communications, distributed MIMO, network modelling, statistical channel modelling and AI-based decision-making. Research interests increasingly extend toward AI-native telecommunications, intelligent network control, AI-assisted network modelling and optimization, hardware-aware AI and adaptive learning for next-generation communication systems. Author/co-author of peer-reviewed research publications in wireless communications and AI-based network optimization.

## RESEARCH INTERESTS

- 6G radio concepts & future wireless network architectures
- AI-native 5G/6G Networks
- Massive MIMO, Cell-Free Massive MIMO & advanced antenna systems
- AI/ML for wireless systems: scheduling, mobility, beam management, interference coordination
- mmWave & Sub-THz communications
- Intelligent radio resource management & energy-efficient network optimization
- Integrated Sensing and Communications (ISAC)
- System-level simulation & performance evaluation
- Reinforcement Learning

## EXPERIENCE

### Researcher experience

04/2023 – 08/2026

**Pan African University for Basic Science, Technology and Innovation (PAUSTI), Nairobi, Kenya**

Research focused on the development of AI-based optimization methods for large-scale wireless networks.

#### Key research activities:

- Developed deep and multi-agent reinforcement learning algorithms for distributed radio resource management, achieving 40% energy-efficiency improvement versus conventional power control baselines.
- Designed and ran system level simulations of uplink and downlink power control and resource under interference in large-scale, distributed MIMO architectures, evaluating up to 200 access points / 45 users.
- Built MATLAB and Python-based (PyTorch) simulation frameworks for algorithm evaluation, performance benchmarking, and reproducible wireless system studies used across all four published studies.
- Investigated mmWave Cell-Free Massive MIMO trade-offs between spectral efficiency, energy efficiency, interference management, and computational complexity, informing algorithm design choices that improved system performance metrics.
- Used AI-assisted tools (LLM-based coding assistants) to accelerate simulation development, debugging, and literature review throughout the research workflow.
- Authored 4 peer-reviewed publications (Web of Science & Scopus) as first author, covering power control, multi-agent RL, and energy-efficient resource allocation for mmWave channel and Sub-6 channel cell-free massive MIMO.

### Part-time Lecturer

09/2023 – 08/2025

**Jomo Kenyatta University of Agriculture and Technology (JKUAT), Nairobi, Kenya**

- Taught signal processing, communication systems and optimization.
- Supervised engineering and research projects.
- Guided students in mathematical modelling, programming and computational methods.
- Applied simulation-based methodologies to engineering and communication-system problems.
- Supported students in developing computational solutions to complex engineering problems.

### Telecommunications Engineer

12/2022 – 04/2023

**IMPERIS Sarl, Bamako, Mali**

Designed and deployed optical-fiber and microwave (RF) communication systems; performed installation, configuration, and technical testing of RF/microwave infrastructure equipment; diagnosed and resolved real-world RF/communication system issues under field conditions.

#### **Technical Intern**

07/2022 – 11/2022

Orange Mali, Bamako, Mali

Supported FTTX optical access network deployment and configuration; contributed to wireless access network implementation and technical documentation.

#### **Part-time Lecturer (multiple institutions)**

2021 – 2024

ISPATEC, INTEC-Sup, EUTG, École Relationnelle d'Informatique et Électronique - Bamako, Mali

Taught analog/digital communication systems, network planning, electronics, digital systems, algorithms and Java programming, and Linux/Windows Server administration; supervised laboratory sessions and hands-on projects.

## **EDUCATION**

---

#### **Ph.D., Electrical Engineering (Telecommunications option)**

04/2023 – 08/2026

Pan African University for Basic Science, Technology and Innovation (PAUSTI)

Thesis: Deep Reinforcement Learning for Energy-Efficient Power Control and Allocation in User-Centric Cell-Free Massive MIMO Networks.

##### **Research focus:**

- Deep Reinforcement Learning for wireless-network optimization
- Multi-Agent Deep Reinforcement Learning for distributed decision-making
- Power control and resource allocation
- Energy-efficient wireless-network optimization
- Large-scale user-centric cell-free massive MIMO
- mmWave wireless communications
- Mathematical modelling and constrained optimization
- Computational simulation and performance evaluation

#### **M.Sc., Electrical Engineering (Telecommunications Systems)**

10/2019 – 07/2021

University 20 August 1955, Skikda

Thesis: Design of a Dual-Band 28/38 GHz MIMO Antenna with High Isolation for 5G Applications Using HFSS Simulation.

##### **Research included:**

- Electromagnetic modelling
- 28/38 GHz mmWave antenna design
- MIMO systems
- S-parameter analysis
- Coupling and isolation optimization
- HFSS-based simulation
- RF performance analysis

#### **B.Sc., Telecommunications**

10/2016 – 07/2019

University 20 August 1955, Skikda

## **SKILLS**

---

- **Wireless Communications:** 6G radio concepts, Cell-Free Massive MIMO, Massive MIMO, mmWave/Sub-THz communications, advanced antenna systems, radio resource management, scheduling, beam management, interference coordination, power control, energy-efficient networks
- **AI & Machine Learning:** Reinforcement Learning, Deep Reinforcement Learning, Multi-Agent RL, Deep Learning, PyTorch, TensorFlow, LLM, Gen AI, AI-assisted research workflows and tooling
- **Signal Processing & Optimization:** Stochastic modeling, numerical optimization, channel modeling, interference management, system-level performance analysis
- **Programming & Simulation:** Python, MATLAB, Java, C++, R, SQL; system-level simulation, data analysis, algorithm development and automation
- **Antenna & RF Engineering:** Ansys HFSS, CST; MIMO antenna design, electromagnetic simulation, S-parameter and radiation analysis
- **Data & Systems:** Data analysis and visualization, Linux/Unix, AWS, Azure, shell scripting
- Languages: English (fluent), French (native)

## **LANGUAGES**

---

- **English:** fluent

- **French: native**

## RESEARCH PUBLICATIONS

---

- D. Diarra, H. O. Absaloms, and P. K. Langat, "Deep reinforcement learning-based power control optimization for maximizing energy efficiency in mmWave user-centric cell-free massive MIMO," IET Communications, vol. 20, 2026. DOI: <https://doi.org/10.1049/cmu2.70199>
- D. Diarra, H. O. Absaloms, and P. K. Langat, "Multi-agent deep reinforcement learning-based power control for energy-efficient user-centric cell-free massive MIMO," Physical Communication, vol. 79, 2026. DOI: <https://doi.org/10.1016/j.phycom.2026.103275>
- D. Diarra, H. O. Absaloms, and P. K. Langat, "Optimizing uplink power control for energy efficiency in mmWave user-centric cell-free massive MIMO with deep reinforcement learning," Journal of Engineering and Applied Science, vol. 73, 2026. DOI: <https://doi.org/10.1186/s44147-025-00864-w>
- D. Diarra, H. O. Absaloms, and P. K. Langat, "Deep reinforcement learning-based power allocation for maximizing energy efficiency in mmWave cell-free massive MIMO," International Journal of Intelligent Engineering and Systems, vol. 18, 2025. DOI: <https://doi.org/10.22266/ijies2025.1130.29>
- D. Diarra, H. O. Absaloms, and P. K. Langat, "Energy-efficient downlink power allocation in user-centric cell-free massive MIMO networks using hybrid deep reinforcement learning algorithm," IEEE Transactions on Communications, under review.

## AWARDS & CERTIFICATIONS

---

- |                                                                                |             |
|--------------------------------------------------------------------------------|-------------|
| ▪ Learning AI Through Visualization, Columbia University                       | <b>2025</b> |
| ▪ Pan African University (PAUSTI) Scholarship                                  | <b>2023</b> |
| ▪ Algeria-Mali Cooperation Scholarship (Excellence Award)                      | <b>2016</b> |
| ▪ English Language Level C, PAUSTI 2023                                        |             |
| ▪ English Language Certification (B2 – Independent User), University of Skikda | <b>2020</b> |